difference in entropy between their ordered and disordered configurations. And this means that there is a huge likelihood that the systems will evolve from lower to higher entropy, which is a rough statement of the second law of thermodynamics. But the key fact to notice is that the second law is *derivative:* it is merely a consequence of probabilistic reasoning applied to Newton's laws of motion.

This leads us to a simple but astounding point: *Since Newton's laws of physics have no built-in temporal orientation, all of the reasoning we have used to argue that systems will evolve from lower to higher entropy toward the future works equally well when applied toward the past.* Again, since the underlying laws of physics are time-reversal symmetric, there is no way for them even to distinguish between what we call the past and what we call the future. Just as there are no signposts in the deep darkness of empty space that declare this direction up and that direction down, there is nothing in the laws of classical physics that says this direction is time future and that direction is time past. The laws offer no temporal orientation; it's a distinction to which they are completely insensitive. And since the laws of motion are responsible for how things change—both toward what we call the future and toward what we call the past—the statistical/probabilistic reasoning behind the second law of thermodynamics applies equally well in both temporal directions. Thus, *not only is there an overwhelming probabilitythat the entropy of a physical system will be higher in what we call the future, but there is the same overwhelming probability that it was higher in what we call the past.* We illustrate this in Figure 6.2.

This is *the* key point for all that follows, but it's also deceptively subtle. A common misconception is that if, according to the second law of thermodynamics, entropy increases toward the future, then entropy necessarily *decreases* toward the past. But that's where the subtlety comes in. The second law actually says that if at any given moment of interest, a physical system happens not to possess the maximum possible entropy, it is extraordinarily likely that the physical system will subsequently have *and* previously had more entropy. That's the content of Figure 6.2b. With laws that are blind to the past-versus-future distinction, such time symmetry is inevitable.